

Q.1 $\underline{A} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$

(a) Find the orthogonal projection matrix onto $\text{Col}(\underline{A})$.

(b) Determine the singular values of $\underline{A}$

A.1 $\underline{A} = [\underline{a}_1 \ \underline{a}_2]$, $\underline{a}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$, $\underline{a}_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$

(a) Apply G-S procedure to find an orthonormal basis for $\text{Span}\{\underline{a}_1, \underline{a}_2\} = \text{Col}(\underline{A})$:

$$\hat{\underline{q}}_1 = \frac{\underline{a}_1}{\|\underline{a}_1\|_2} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad \underline{e}_2 = \underline{a}_2 - \hat{\underline{q}}_1 \hat{\underline{q}}_1^T \underline{a}_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1/2 \\ 1 \\ -1/2 \end{bmatrix}$$

$$\hat{\underline{q}}_2 = \frac{\underline{e}_2}{\|\underline{e}_2\|_2} = \frac{\begin{bmatrix} 1/2 \\ 1 \\ -1/2 \end{bmatrix}}{\sqrt{3/2}} = \sqrt{\frac{2}{3}} \begin{bmatrix} 1/2 \\ 1 \\ -1/2 \end{bmatrix}$$

$$\underline{P} = \hat{\underline{q}}_1 \hat{\underline{q}}_1^T + \hat{\underline{q}}_2 \hat{\underline{q}}_2^T = \frac{1}{2} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \end{bmatrix} + \frac{2}{3} \begin{bmatrix} 1/2 \\ 1 \\ -1/2 \end{bmatrix} \begin{bmatrix} 1/2 & 1 & -1/2 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$$

(b) Singular Values of $\underline{A}$ are the square root of the eigenvalues of $\underline{A}^T \underline{A}$

$$\underline{A}^T \underline{A} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \quad \det(\underline{A}^T \underline{A} - \lambda \underline{I}) = \begin{vmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{vmatrix} = (2-\lambda)^2 - 1$$

Hence the characteristic equation of $\underline{A}^T \underline{A}$ becomes:

$$p(\lambda) = \lambda^2 - 4\lambda + 3 = (\lambda - 3)(\lambda - 1) \Rightarrow \lambda_1 = \sigma_1^2 = 3$$

$$\lambda_2 = \sigma_2^2 = 1$$

Therefore

$$\sigma_1 = \sqrt{3}, \quad \sigma_2 = 1$$

Q.2 Find the best fitting first order polynomial of time to the following set of measurements:

t	0	1	2	3	4
$x(t)$	9	21	28	41	50

A.2 Want to minimize: $\| \underline{x} - \underline{A} \cdot \underline{\alpha} \|_2^2$

where $\underline{x} = [9 \ 21 \ 28 \ 41 \ 50]^T$, $\underline{\alpha} = [\alpha_1 \ \alpha_2]^T$

$$\underline{A} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{bmatrix}, \text{ this } \underline{A} \text{ corresponds to the following model:}$$

$$x(t) = \alpha_1 + \alpha_2 t$$

Then the optimal $\underline{\alpha}$ can be found by using the orthogonality property:

$$\underline{A}^T (\underline{x} - \underline{A} \underline{\alpha}) = \underline{0}$$

$$\Rightarrow \underline{\alpha} = (\underline{A}^T \underline{A})^{-1} \underline{A}^T \underline{x}$$

$$\underline{A}^T \underline{A} = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 & 4 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{bmatrix} = \begin{bmatrix} 5 & 10 \\ 10 & 30 \end{bmatrix}$$

$$(\underline{A}^T \underline{A})^{-1} = \frac{1}{150 - 100} \begin{bmatrix} 30 & -10 \\ -10 & 5 \end{bmatrix} = \frac{1}{50} \begin{bmatrix} 30 & -10 \\ -10 & 5 \end{bmatrix} = \frac{1}{10} \begin{bmatrix} 6 & -2 \\ -2 & 1 \end{bmatrix}$$

$$\underline{A}^T \underline{x} = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 & 4 \end{bmatrix} \begin{bmatrix} 9 \\ 21 \\ 28 \\ 41 \\ 50 \end{bmatrix} = \begin{bmatrix} 149 \\ 400 \end{bmatrix}$$

Hence optimal coefficients are $\begin{bmatrix} \alpha_1 \\ \alpha_2 \end{bmatrix} = \frac{1}{10} \begin{bmatrix} 6 & -2 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} 149 \\ 400 \end{bmatrix} = \begin{bmatrix} 94/10 \\ 102/10 \end{bmatrix}$

Therefore, best fitting polynomial is:

$$\boxed{x(t) = 9.4 + 10.2t}$$